

Dynamic Ecosystem



Tool-based Agents



OpenClaw-Style Orchestrators



Multi-tool Orchestrators

Typed API & Intent Schema

Constrained Transition δ

Compositional
Robustness

Semantic
Containment

Accountability &
Attributability

Cross-Boundary
Reliability

Safe State Space S_{safe}

- ❑ Capability-restricted action schemas
- ❑ Typed state machine orchestration
- ❑ Formal verification of critical flows

- ❑ Typed intent schema
- ❑ Semantic type system
- ❑ Constraint-aware planners

- ❑ State-level provenance encoding
- ❑ Incremental hash chain of transitions
- ❑ Causal graph representation

- ❑ Bounded-depth planning trees
- ❑ Deadlock detection
- ❑ Deterministic termination states